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Advancing FAIR meta-analyses of nucleotide sequence data with *q2-fondue*

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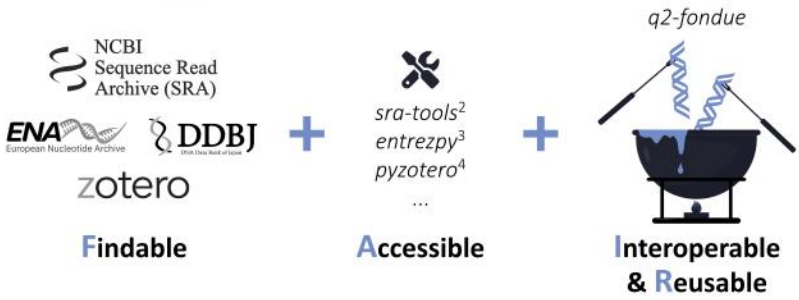
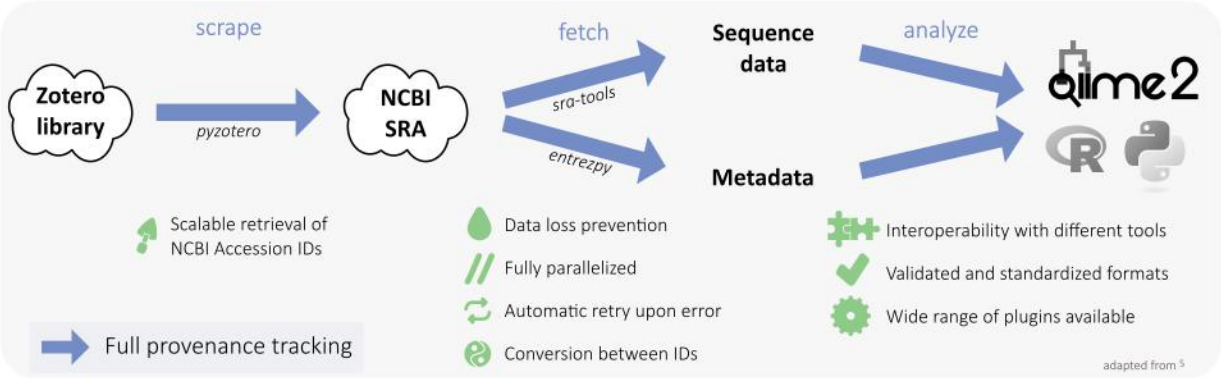
Combining sequence databases with programmatic tools for FAIR meta-analyses

In the past two decades, studies started making their sequencing (meta)data **findable** in their publications and through different databases. To avoid manual data retrieval steps, tools have emerged that make sequence (meta)data programmatically **accessible**. Since the existing tools were developed independently, they do not support **interoperable**, **reusable** workflows capable of tracking every data retrieval and analysis step performed. Also, users with limited programming skills struggle to use them due to complex setup requirements. To overcome these bottlenecks, we developed *q2-fondue*, an open source Python package, that enables a **FAIR**[1] conduct of meta-analyses.

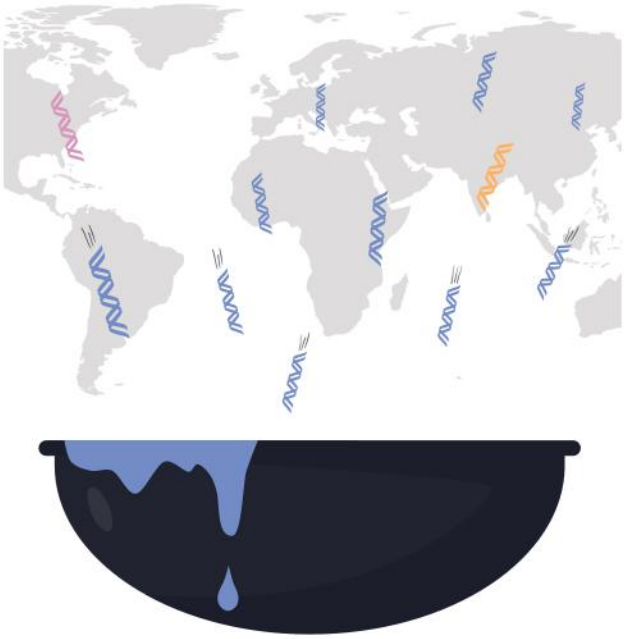


q2-fondue streamlines acquisition & analysis of sequence (meta)data ...



- Open-source
- Continuously maintained
- QIIME 2[6] interoperable

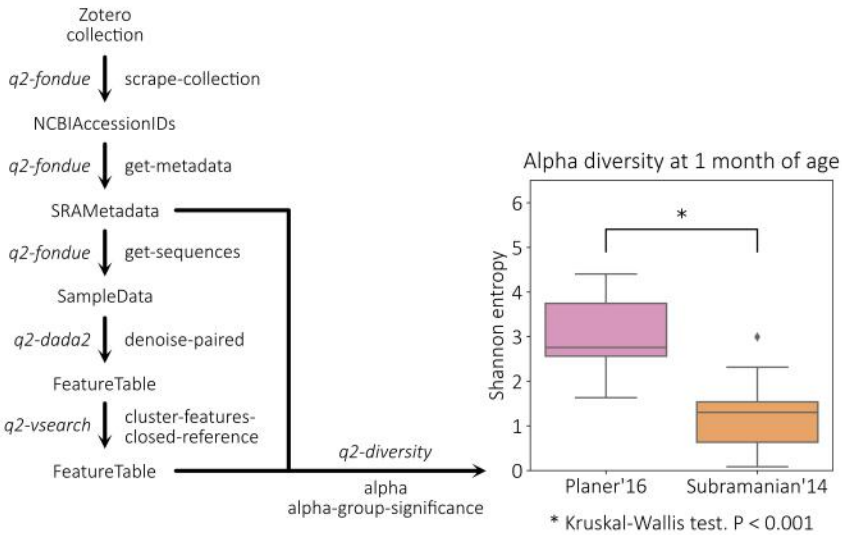
... with full provenance tracking from data download to final visualization in QIIME 2



An example amplicon sequence analysis of the gut microbiome of infants at 1 month of age from two different studies

Planer et al. 2016[7] — Subramanian et al. 2014[8]

using *q2-fondue* and other QIIME 2 packages yields a fully provenance-tracked visualization:



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Publication **Software**